Crime Detection Datasets

Overview & Bibliographic References

RWF-2000

Description	Large-scale video dataset for real-world violence detection, collected from surveillance cameras and online sources.
Videos	2,000
Classes	2 **Fighting** | **Non-Fighting**
Distribution	1,000 videos per class (balanced)
Size	12.7 GB
Split	Train / Test
Source	Kaggle
Link	https://www.kaggle.com/datasets/daniazehra/rwf-2000

Bibliographic Reference

Cheng, M., Cai, K., & Li, M. (2021). *RWF-2000: An Open Large Scale Video Database for Violence Detection.* In *Proceedings of the 25th International Conference on Pattern Recognition (ICPR 2020)*, pp. 4183–4190. IEEE.

DOI: 10.1109/ICPR48806.2021.9412502

BibTeX:

```
@InProceedings{cheng2021rwf,
   author = {Cheng, Ming and Cai, Kunjing and Li, Ming},
   title = {{RWF-2000}:  An Open Large Scale Video Database for Violence Detection},
   booktitle = {Proc. IEEE ICPR},
   year = {2021},
   pages = {4183-4190},
   doi = {10.1109/ICPR48806.2021.9412502}
}
```

UCF Crime Full

Description	Large-scale dataset for anomaly detection in surveillance videos, covering 13 real-world crime categories plus normal footage. Originally proposed as part of the UCF-Crime benchmark.
Videos	1,000 (subset) — Full dataset: 1,900 videos
Size	26.8 GB
Source	Kaggle
Link	https://www.kaggle.com/datasets/alirakhmaev/ucf-crime-full

Classes (14):

01. Abuse	06. Explosion	11. Shooting
02. Arrest	07. Fighting	12. Shoplifting
03. Arson	08. Normal Videos	13. Stealing
04. Assault	09. Road Accidents	14. Vandalism
05. Burglary	10. Robbery	

Bibliographic Reference

Sultani, W., Chen, C., & Shah, M. (2018). *Real-World Anomaly Detection in Surveillance Videos.* In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2018)*, pp. 6479–6488.

DOI: 10.1109/CVPR.2018.00678 **arXiv:** 1801.04264

BibTeX:

```
@InProceedings{sultani2018real,
    author = {Sultani, Waqas and Chen, Chen and Shah, Mubarak},
    title = {Real-World Anomaly Detection in Surveillance Videos},
    booktitle = {Proc. IEEE/CVF CVPR},
    year = {2018},
    pages = {6479-6488},
    doi = {10.1109/CVPR.2018.00678}
}
```

CamNuvem — A Robbery Dataset for Video Anomaly Detection

Description	Brazilian surveillance video dataset focused on robbery detection, designed for weakly-supervised anomaly detection research. Videos collected from real CCTV footage.
Videos	972
Classes	2 **Abnormal** (robbery events) | **Normal**
Size	6.57 GB
Source	GitHub
Link	https://github.com/daviduarte/camnuvem-dataset

Bibliographic Reference

de Paula, D. D., Salvadeo, D. H. P., & de Araujo, D. M. N. (2022). *CamNuvem: A Robbery Dataset for Video Anomaly Detection. Sensors*, 22(24), Article 10016. MDPI.

DOI: 10.3390/s222410016 **PubMed ID:** 36560385 **ISSN:** 1424-8220

URL: https://www.mdpi.com/1424-8220/22/24/10016

BibTeX:

```
@Article{depaula2022camnuvem,
    author = {de Paula, Davi D. and Salvadeo, Denis H. P.
          and de Araujo, Darlan M. N.},
    title = {{CamNuvem}:  A Robbery Dataset for Video Anomaly Detection},
    journal = {Sensors},
    volume = {22},
    number = {24},
    article-number = {10016},
    year = {2022},
    issn = {1424-8220},
    doi = {10.3390/s222410016},
```

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    pubmedid = {36560385}
}
```

Comparative Summary

Dataset	Videos	Classes	Size	Task
RWF-2000	2,000	2	12.7 GB	Binary violence detection
UCF Crime Full	1,000	14	26.8 GB	Multi-class anomaly detection
CamNuvem	972	2	6.57 GB	Binary robbery detection
Total	**3,972**	**—**	**≈46 GB**	

All three datasets are publicly available and were used in the unified corpus of 2,974 videos after class mapping and filtering (RWF-2000 [1], UCF-Crime [2], CamNuvem [3]).

References

[1] M. Cheng, K. Cai, and M. Li, “RWF-2000: An open large scale video database for violence detection,” in *Proc. IEEE ICPR*, 2021, pp. 4183–4190.

[2] W. Sultani, C. Chen, and M. Shah, “Real-world anomaly detection in surveillance videos,” in *Proc. IEEE/CVF CVPR*, 2018, pp. 6479–6488.

[3] D. D. de Paula, D. H. P. Salvadeo, and D. M. N. de Araujo, “CamNuvem: A robbery dataset for video anomaly detection,” *Sensors*, vol. 22, no. 24, art. 10016, 2022, doi: 10.3390/s222410016.